GG-K-00508c(DSA-DM)
8 January 1969
Interim revision of
Fed. Spec. GG-K-508b
27 March 1957

INTERIM FEDERAL SPECIFICATION

KNIFE, PLASTER, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-K-508b.

* 1. SCOPE

* 1.1 Scope. This specification covers a knife suitable for dental use in trimming plaster and in separating sections of dental flasks.

* 2. APPLICABLE DOCUMENTS

* 2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

FSC 6520

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STANDARD

Federal

FED-STD-151 Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATION

Military

MIL-P-116 Preservation, Methods of.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM Designation E 92	Vickers Hardness of Metallic Materials.
ASTM Designation E 140	Standard Hardness Conversion Tables for Metals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material.

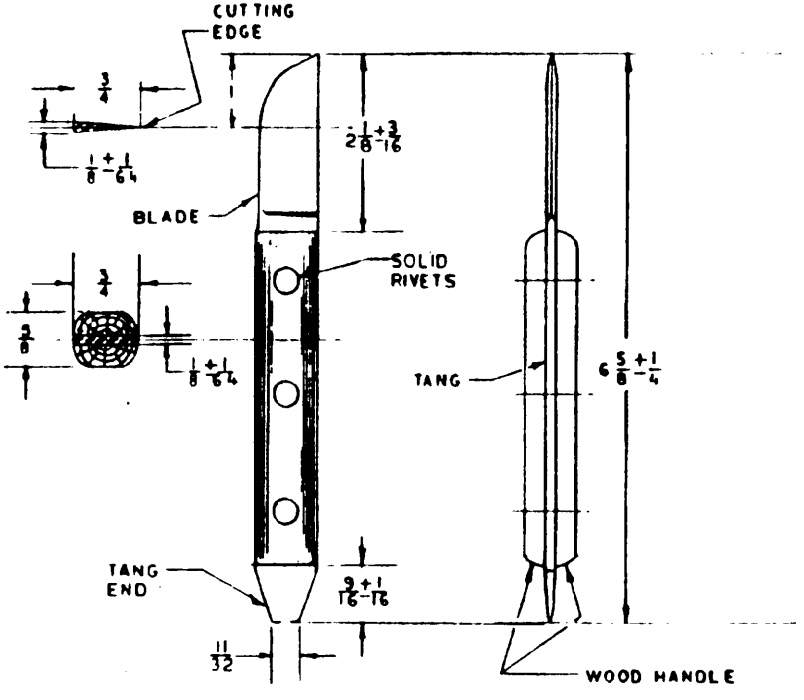
* 3.1.1 Blade and tang. The blade and tang shall be fabricated from carbon steel containing 0.90 to 1.10 percent carbon.

* 3.1.2 Handle. The wood used to construct the handle shall be fabricated from the same species of solid hardwood.

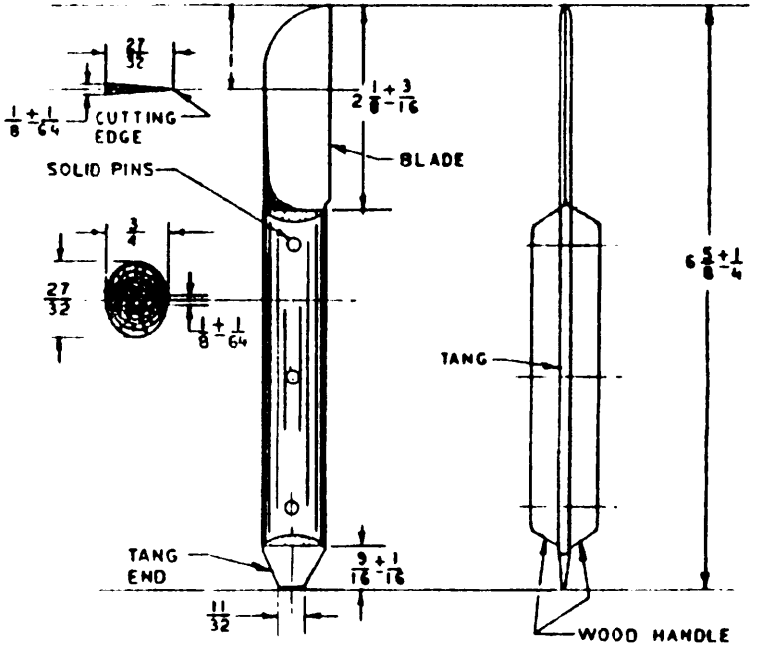
* 3.1.3 Rivets or pins. The rivets or pins, as applicable, shall be fabricated from brass or from nickel silver.

* 3.2 Style, design, and dimensions. The plaster knife shall be of the Gritman design. The knife shall consist of a blade and tang, and a two-piece hardwood handle secured with a minimum of three rivets or pins. The knife shall be in accordance with figure 1 for style, design, and dimensions. The blade shall have a straight cutting edge suitable for trimming casts and models of plaster of paris, artificial stone, and dental investment. The tang shall taper to a chisel-type proximal end suitable for separating sections of dental flasks.

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ALTERNATE DESIGN AND DIMENSIONS

Unless otherwise specified, dimensions are in inches and tolerances are ± 1/32.

Figure 1

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* 3.3 Construction. The knife shall be symmetrically made. The handle shall be securely attached to the tang. The rivets or pins shall be blended with both sides of the handle. The pins shall be properly set so as not to permit any perceptible lateral play between the handle and tang during use. The sides of the tang shall be tapered toward or blended with both sides of the hardwood handle, and shall not dig into the hand of the user.

* 3.4 Finish. The finish on the metal edges and surfaces shall be uniform. The blade and rivets or pins shall have a smooth satin, a smooth crocus, or a bright polished finish. The blade shall be uniformly taper ground and shall be honed to a proper cutting edge for immediate dental use. The handle shall be given a durable coating of varnish or lacquer to impart a smooth finish and to minimize possible moisture absorption.

* 3.5 Hardness. After proper stress relieving, the blade shall have a Rockwell hardness of C58 to C62 (equivalent to a Vickers hardness of 653 to 746, as shown in ASTM Designation E 140), when tested as specified in 4.4.2.

* 3.6 Cutting ability. The cutting edge of the knife shall not dull, turn over, nor break, when tested as specified in 4.4.3.

* 3.7 Identification marking. The blade of each knife shall be permanently and legibly marked with the name or trademark of the manufacturer or supplier.

* 3.8 Workmanship. Workmanship shall be first class throughout. The knives shall be free from defects which detract from their appearance or may impair their serviceability.

* 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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* 4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

* 4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

* 4.2 Quality conformance inspection. The examinations required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and III (see 4.2.1 and 4.3), and the tests required for acceptance are specified in table II (see 4.2.2).

* 4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one knife.

Table I. Examination criteria

Type of defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-1	1.5
Minor A	II	2.5
Minor B	S-1	4.0

* 4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one knife.

Table II. Test criteria

Characteristic	Requirement	Test procedure	Inspection level
Chemical analysis	3.1	4.4.1	See 4.2.3
Hardness	3.5	4.4.2	S-2
Cutting ability	3.6	4.4.3	S-2

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* 4.2.3 Sampling for chemical analysis. Sufficient amount of each lot or batch of raw material used in the construction of the blade and tang shall be selected for chemical analysis.

* 4.2.4 Sample unit usage. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

* 4.3 Examination. The knives shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with, but shall not be restricted to the classification of defects listed in table III.

Table III. Classification of defects

Category	Defect
Major A:	
101	Handle not securely attached to tang.
102	Style and design of knife not as specified.
103	Cutting edge of blade not properly honed, not ready for use.
104	One or more rivets or pins missing.
105	Perceptible lateral play or wobbling between handle and tang.
106	Blade or tang chipped, cracked, or broken.
107	Cutting edge of blade not free from defects visible to the unaided eye such as breaks, jags, or nicks.
108	Handle not free from workmanship defects such as slivers, rough edges, loose grains, knots or splits.
Major B:	
151	Dimensions of blade and chisel end of knife not as specified.
Minor A:	
201	Rivets or pins not flush with outer surface of wooden handle.
202	Sides of tang do not blend with sides of handle.
203	Handle not properly lacquered or varnished.
204	Chisel end of knife bent or twisted.
205	Blade and tang not finished smooth; not free from workmanship defects such as burrs, dents, sharp edges, gouges, or pits.
206	Blade not permanently and legibly marked with name or trademark of manufacturer or contractor.
207	Packaging not as specified.
208	Packing not as specified.
209	Marking on packaging and pack not as specified.
Minor B:	
251	Dimensions of handle not as specified.

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* 4.4 Test procedures.

* 4.4.1 Chemical analysis of raw material. In the absence of certificates of quality, the chemical composition of the steel used in the manufacture of the knives shall be determined using method 111 or 112 of FED-STD-151.

* 4.4.2 Hardness. The hardness of the blade shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers). The hardness readings shall be taken on the area immediately adjacent to the cutting edge of the blade and shall not extend onto any beveled portion of the edge.

* 4.4.3 Cutting ability. The blade of each knife shall be tested by being used ten times to trim a cast, a model of plaster of paris, or artificial stone.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One knife, as specified, constitutes one unit of issue.

* 5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	360 units

* 5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

* 5.1.3 Level A.

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* 5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

* 5.1.3.2 Preservative. Each unit shall be preserved with type P-9 preservative in accordance with MIL-P-116.

* 5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

* 5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-15.

* 5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with either adhesive (conforming to MMM-A-250) or metal stitches is permitted.

* 5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

* 5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

* 5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table III, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

* 5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table II, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

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* 5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

* 5.3 Marking.

* 5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

* 5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type which is normally procured to meet Federal Government requirements.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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Review activities:
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Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION	FSN
GG-K-00508c(DSA-DM)/	

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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